

```

void arith32 (
    int A,    int B,
    int C,    int D,
    int *out1, int *out2,
    int *out3
) {
    // Basic arithmetic operations
    *out1 = A + B;
    *out2 = A - B;
    *out3 = C * D;
}

```

```

void arith20 (
    ap_int<20> A,    ap_int<20> B,
    ap_int<20> C,    ap_int<20> D,
    ap_int<20> *out1, ap_int<20> *out2,
    ap_int<20> *out3
) {
    // Basic arithmetic operations
    *out1 = A + B;
    *out2 = A - B;
    *out3 = C * D
}

```

